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Kate Sazonova

Sunday, October 18, 2026

Smoking Addiction Dataset

This is a synthetic longitudinal behavioral dataset, available on Kaggle, and containing simulated observations of nicotine-use behaviors for 1000 participants over the course of 30 days, along with observations of their psychological states and daily habits like sleep, exercise or caffeine consumption. The dataset has a Usability rating of 10.0 and contains 35 columns and 30,000 records total.

The dataset contains the following features, which here I separated into thematic categories for better interpretability:

Record identifiers:

- Participant_ID
- Day_Number

Participant Demographics:

- Age_Group
- Gender
- Occupation_Status

Nicotine-Use and Craving Behaviors:

- Nicotine_Use_Duration_Months
- Pouches_Per_Day
- Pouch_Strength_mg
- First_Pouch_Time
- Time_Since_Last_Pouch_Minutes
- Longest_Nicotine_Free_Period_Hours
- Morning_Cravings_Score
- Afternoon_Cravings_Score
- Evening_Cravings_Score
- Overall_Craving_Score
- Craving_Triggers
- Craving_Intensity_Peak
- Craving_Duration_Minutes
- Attempted_To_Resist_Craving

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- Successfully_Resisted

Psychological State Metrics:

- Daily_Anxiety_Score
- Morning_Anxiety_Score
- Evening_Anxiety_Score
- Perceived_Stress_Score
- Mood_Score
- Irritability_Score
- Restlessness_Score

Physical and Lifestyle Metrics:

- Sleep_Duration_Hours
- Sleep_Quality_Score
- Caffeine_Intake_mg
- Exercise_Minutes
- Screen_Time_Hours
- Work_Or_Study_Hours
- Major_Stress_Event_Today
- Social_Interaction_Hours

Shown below is a sample view of the dataset, showing first five-day observation entries in a participant, with their respective features:

Participant 1 (P0001):

Participant...	Day_Numb...	Age_Group	Gender	Occupatio...	Nicotine_U...	Pouches_P...	Pouch_Str...	First_Pouc...	Time_Sinc...
P0001	1	18-24	Male	Employed	26	4	6	05:15	72
P0001	2	18-24	Male	Employed	26	5	12	05:30	38
P0001	3	18-24	Male	Employed	26	4	4	05:45	99
P0001	4	18-24	Male	Employed	26	6	8	09:45	53
P0001	5	18-24	Male	Employed	26	4	12	08:30	75

Longest_Ni...	Morning_C...	Afternoon_...	Evening_Cr...	Overall_Cr...	Craving_Tri...	Craving_In...	Craving_D...	Attempted...	Successfull...
6.1	3.7	4.4	4	4.5	Morning	3.9	49	No	No
1	3.8	4.8	4.2	3.9	Stress	4.4	34	Yes	No
9.7	4.6	5.6	4.7	5.1	Alcohol	5.5	51	Yes	No
7.1	3.5	3.9	3.7	3.7	Study	4.2	42	No	No
3.6		4	3.7	3.9	Boredom	3.8	33	No	No
6.9	4.9	6.8	5.4	5.8	Habit	7.2	61	Yes	No

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# Daily_Anxie...	# Morning_A...	# Evening_A...	# Perceived_...	# Mood_Score	# Irritability_...	# Restlessne...	# Sleep_Dur...	# Sleep_Qual...	# Caffeine_In...
7.5	7.9	7.4	0.8	7.2	5.7	4.8	7.2	5.7	30
8.9	8.3	9.2	3.3	6.2	5.5	5.6	7	7	200
9	9	8.5	3.3	6	6	5.5	6.1	5.4	190
8.5	9.3	8.2	1.4	7	5.3	5.5	6.8	6.3	250
8.4	8.6	9.5	3.5	7.1	7.4	6.3	5.4	5.7	390

# Exercise_M...	# Screen_Ti...	# Work_Or_S...	✓ Major_Stre...	# Social_Inte...
4	2.7	8.1	No	2.5
15	7.3	14.9	No	0.7
25	5.4	7.2	No	1.7
10	1.2	9.3	No	1.9
29	1.3	7.4	No	4.7

As can be seen in the dataset, there are a number of numerical features representing scores of the participants' different psychological states, such as nicotine craving, anxiety, stress, mood, irritability, and restlessness. These features are suitable for regression tasks, for example, predicting a nicotine craving score or an anxiety score for a new participant based on selected features.

It is important to note that in the dataset, there are 30 recorded observations for each of the 1000 participants, so it must be accounted for when creating train and test splits for classification or regression tasks on new observations. Since they're not 30,000 individual participant records, day entries per participant (30) will have to be included in the same train/test partition.

The dataset consists of synthetically programmed data, so it's important to understand that any conclusions drawn from the data represent simulated relationships, and are not representative of real-world data about nicotine use and related psychological states.

There are a couple of existing learning model implementations for this dataset that I found interesting and helpful, largely due to the rich exploratory analysis performed on the data, as well as examples of different machine learning models trained for regression and classification tasks on nicotine cravings and/or anxiety scores, which I myself also found to be the most interesting features to investigate in this dataset.

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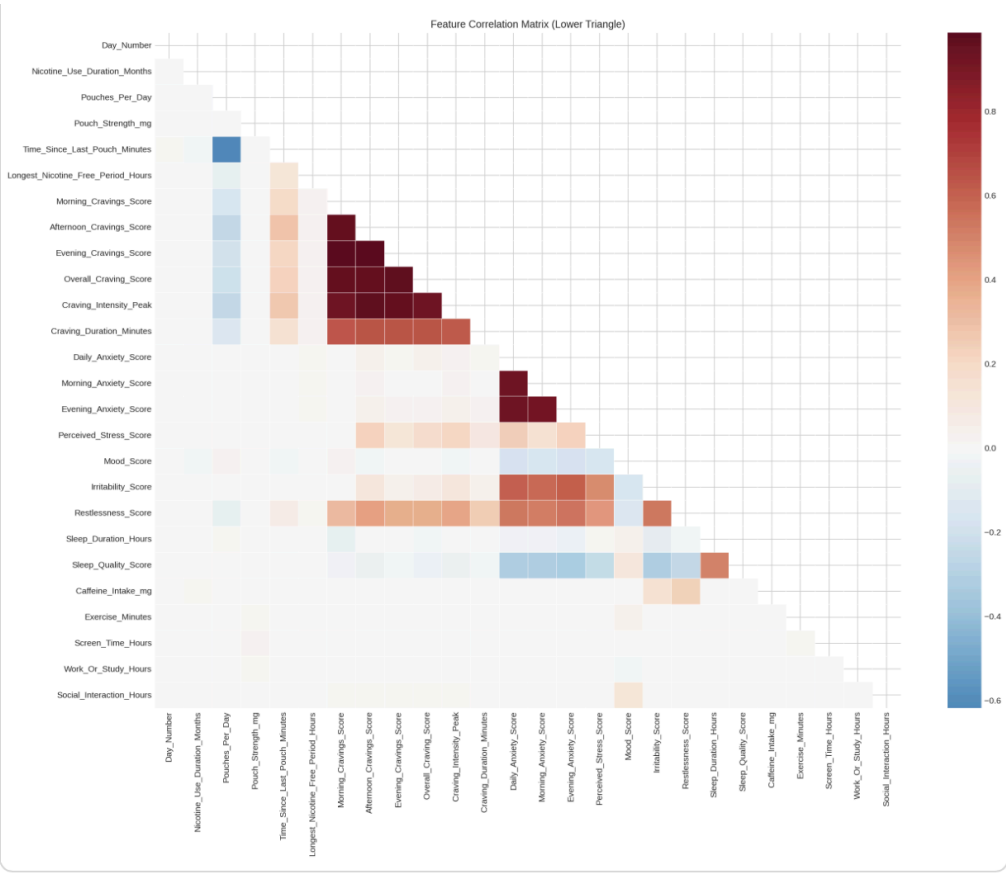
Time-Series Forecasting: Nicotine Craving and Anxiety Dynamics

Kaggle notebook, authored by Rizki Surya Nughoro

This project presents two regression learning tasks: predicting the numerical values of *Overall_Craving_Score* and *Daily_Anxiety_Score* in new participant observations, based on psychological and behavioral measurements, as well as past history of nicotine use. Here, multiple machine learning algorithms are compared to see which one achieves the most accurate prediction of nicotine craving and anxiety scores.

The project first performs a rich exploratory data analysis, which is helpful for understanding the general landscape of the data before the models are trained. Some of the key EDA contributions are shown below.

Feature Correlation Heatmap:



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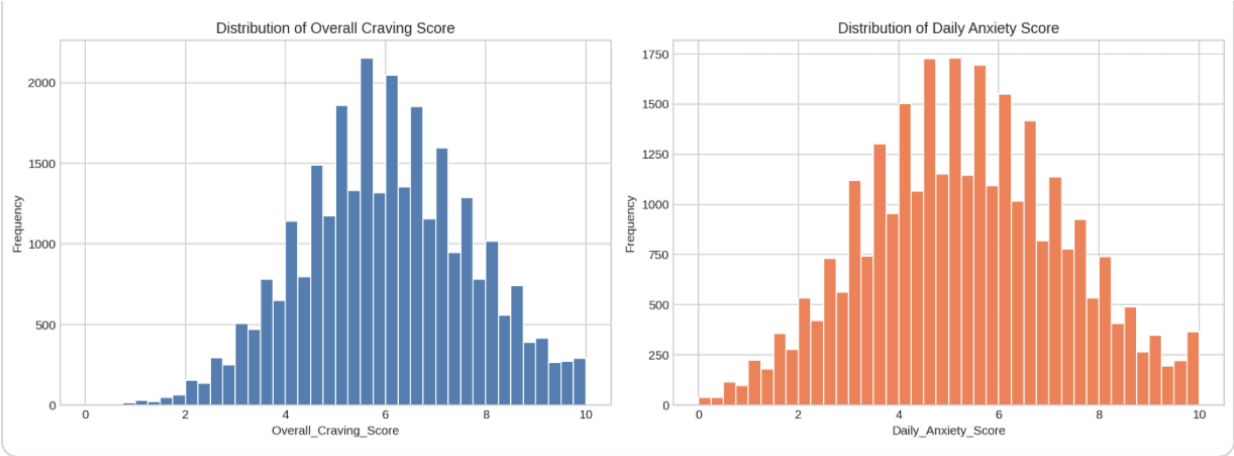
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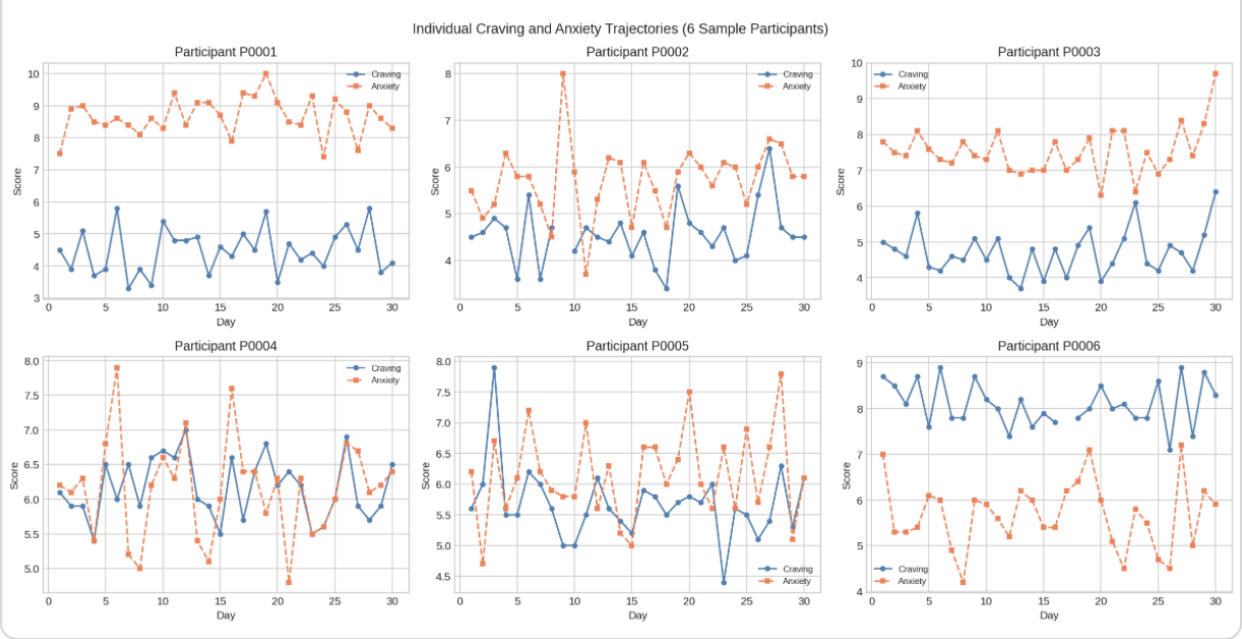
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Target feature distributions:



Feature trajectories over time:



Residual analysis:

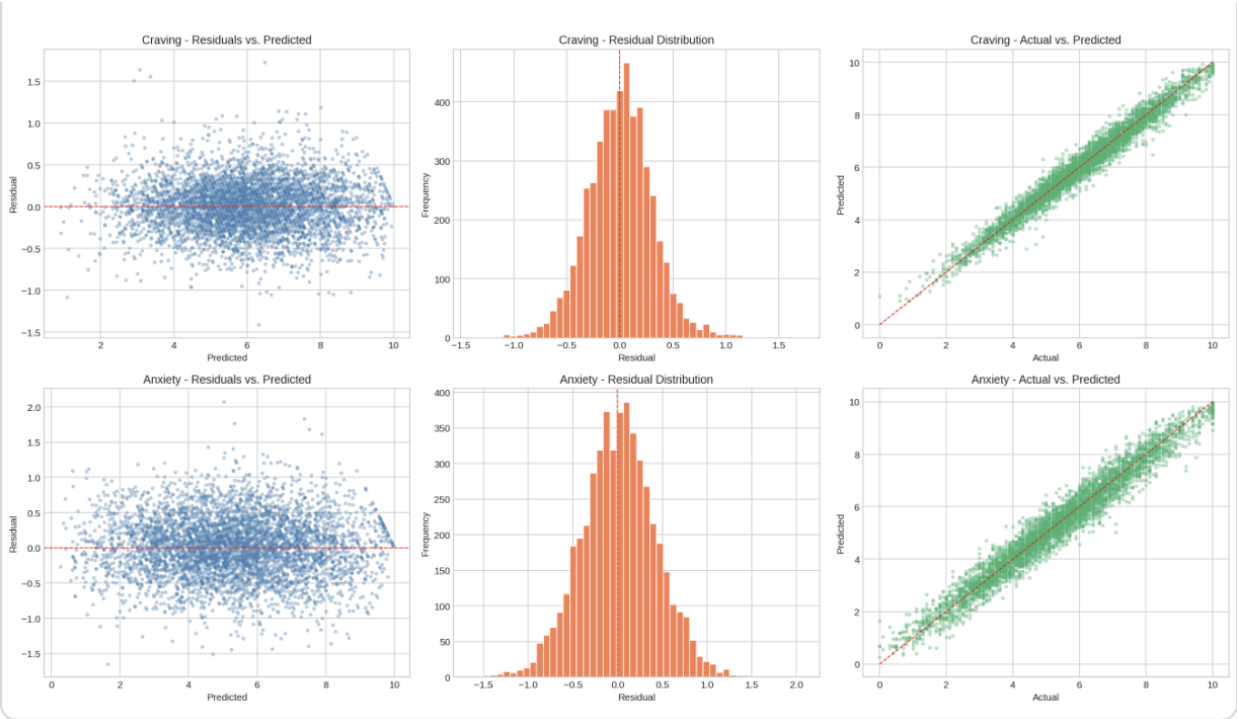
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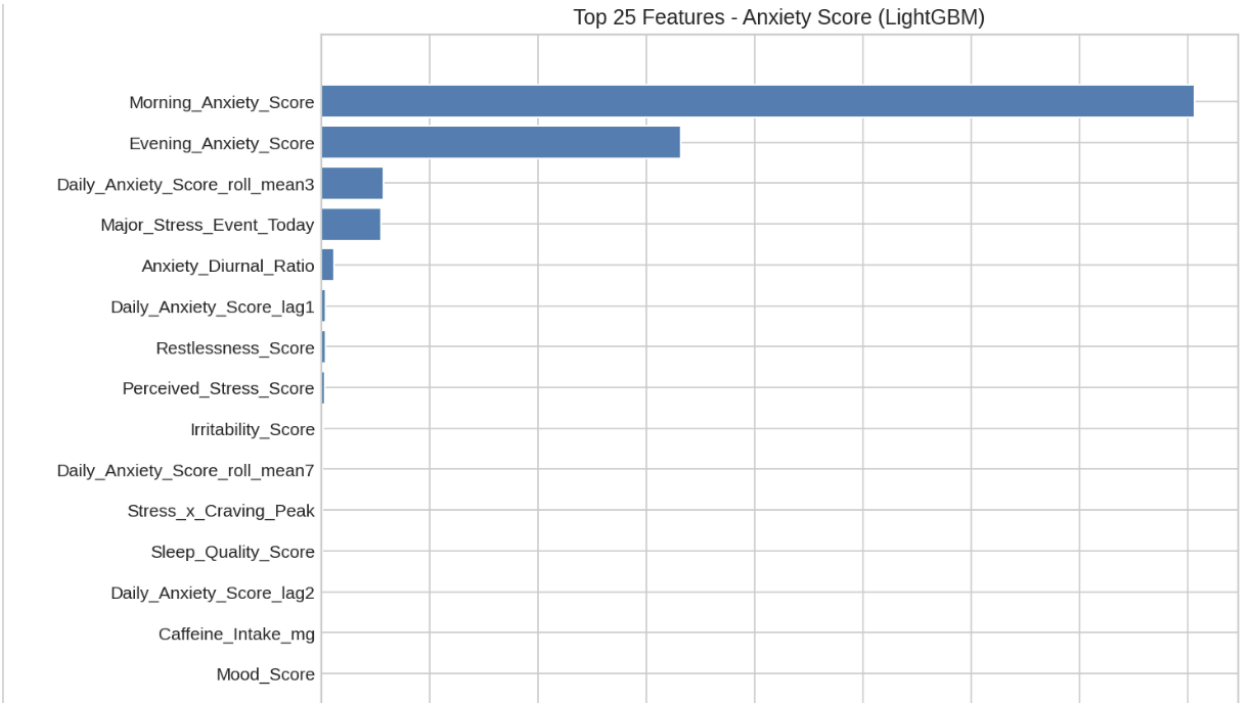
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Additionally, feature importance was evaluated using LightGBM:



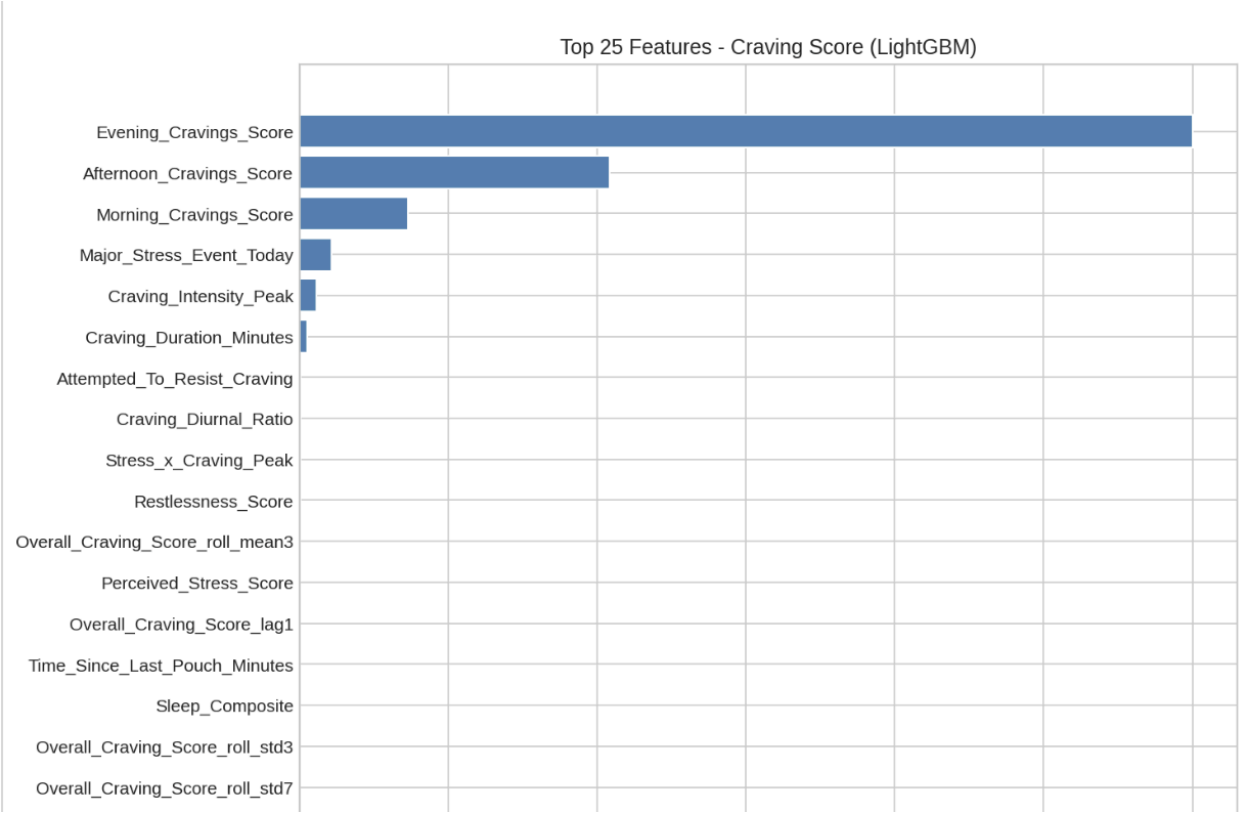
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Data preprocessing for this project involves imputation of missing values, categorical encoding, scaling and feature engineering for previous-day measurements and rolling averages, which was particularly interesting as an example of feature engineering for potential future work on this dataset.

The data here is split chronologically into a training set for days 1-20, validation set for days 21-25, and test set for days 26-30. Cross-validation is performed using a *Time Series Split*.

The models include *Ridge*, *Lasso*, *Elastic Net*, *Random Forest*, *Gradient Boosting*, *Histogram Gradient Boosting*, *XGBoost*, *LightGBM*, and *LSTM*. These models are evaluated against the persistence baseline that predicts the current score using the previous day's score. Evaluation is done using *MAE*, *RMSE*, *MAPE* and R^2.

The evaluation results are as follows:

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Nicotine craving scores:

← **results_craving.csv** (585 B)

Model	Split	MAE	RMSE	MAPE	R2
XGBoost	Test	0.2352	0.3029	78523452876035.8	0.9678
GradientBoosting	Test	0.2362	0.3036	86988131025052.77	0.9677
HistGradientBoosting	Test	0.2367	0.3043	99531457817599.55	0.9676
LightGBM	Test	0.2359	0.3045	98692906847062.5	0.9675
RandomForest	Test	0.2379	0.307	73504522221400.27	0.967
Ridge	Test	0.2566	0.3272	14171786529502.842	0.9625
ElasticNet	Test	0.2567	0.3275	23790624509474.113	0.9624
Lasso	Test	0.2567	0.3281	23824432608621.71	0.9623
LSTM	Test	0.3057	0.3912	147549735547924.28	0.9464
Persistence Baseline	Test	0.7019	0.8991	109621086264609.3	0.7167

Anxiety scores:

← **results_anxiety.csv** (586 B)

Model	Split	MAE	RMSE	MAPE	R2
LightGBM	Test	0.3315	0.4216	296376251019461.6	0.9546
XGBoost	Test	0.3314	0.4221	288158928297009.0	0.9545
GradientBoosting	Test	0.331	0.4221	272599530374037.5	0.9545
HistGradientBoosting	Test	0.3322	0.423	299860353224160.75	0.9543
Ridge	Test	0.3308	0.4257	348767727609230.9	0.9537
ElasticNet	Test	0.3312	0.4259	356776361095468.94	0.9536
Lasso	Test	0.3325	0.4276	368332156756716.2	0.9533
RandomForest	Test	0.339	0.4309	301949762394518.8	0.9525
Persistence Baseline	Test	0.8736	1.1237	261208778387510.06	0.6773
LSTM	Test	1.5246	1.8158	475571946898098.1	0.1575

As seen above, XGBoost achieves the best prediction for craving scores, with RMSE = 0.3029, and R^2 = 0.9678. LightGBM has the lowest anxiety RMSE of 0.4216, and R^2 of 0.9546.

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I think this project is useful for the EDA it provides, the implementation of quite a wide range of different learning models (with provided code for each, that I can later use as an example), and the example of train, validation and test splits being done on the data for each separate participant, rather than the entire split, since that's the kind of split I would have to perform for my future project, as well.

Smoking Addiction Insights

Kaggle notebook, authored by Shair Khan

This project implements three machine learning tasks: regression to predict the numerical value of *Craving_Intensity_Peak*, binary classification to predict whether a participant's daily anxiety score is above the median, and unsupervised clustering that identifies groups of participant observations with similar behavioral characteristics.

The exploratory data analysis here provides some helpful information about the features and their relationships, before the models are trained. It's different from the first project and introduces some more ways to analyze features and feature relationships/dependencies.

Correlation Heatmap:

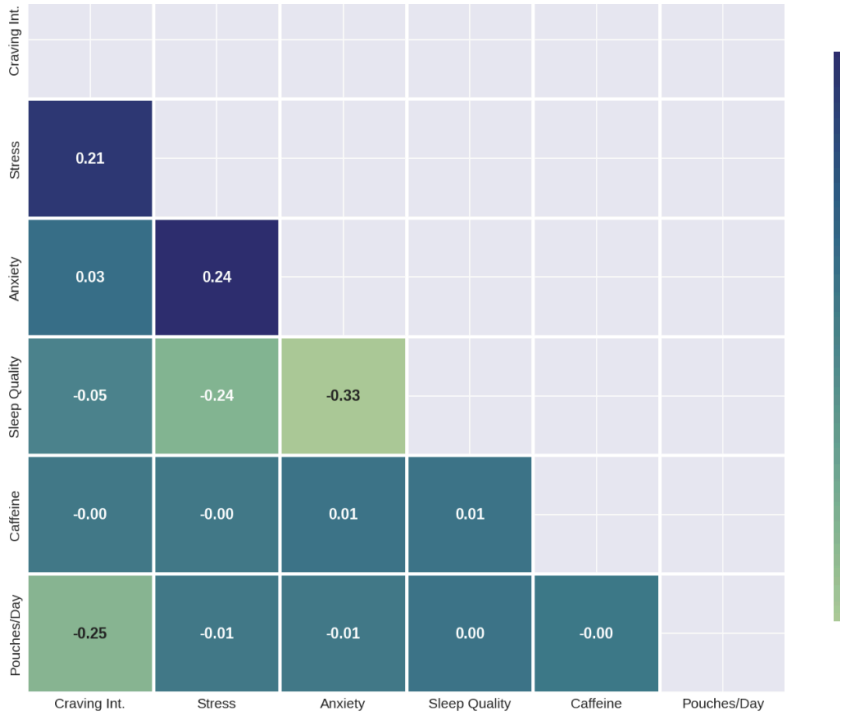
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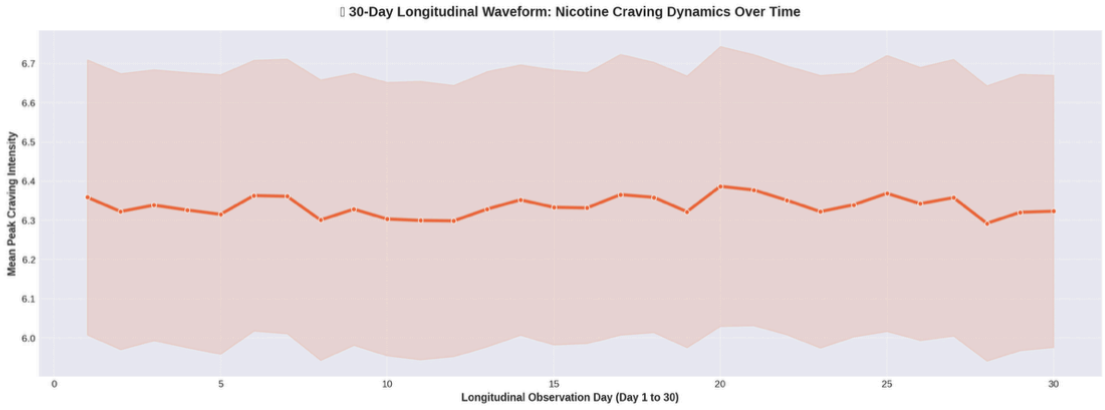
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Longitudinal Waveform (nicotine craving dynamics over time)



Bivariate Density Map (perceived stress vs. composite anxiety)

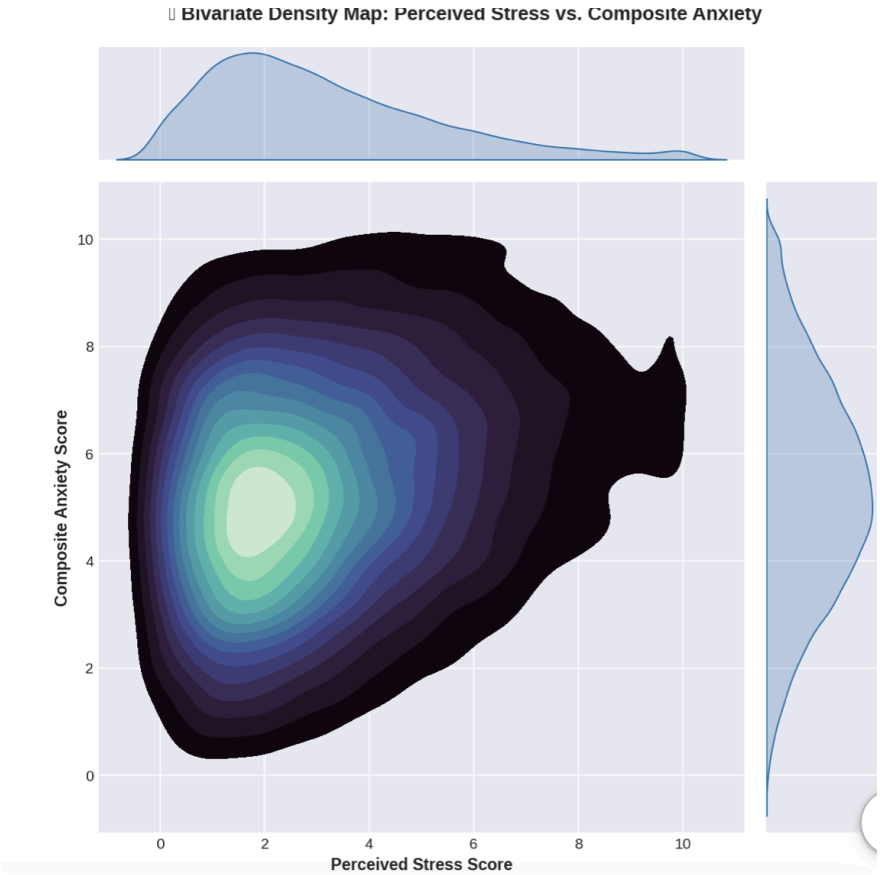
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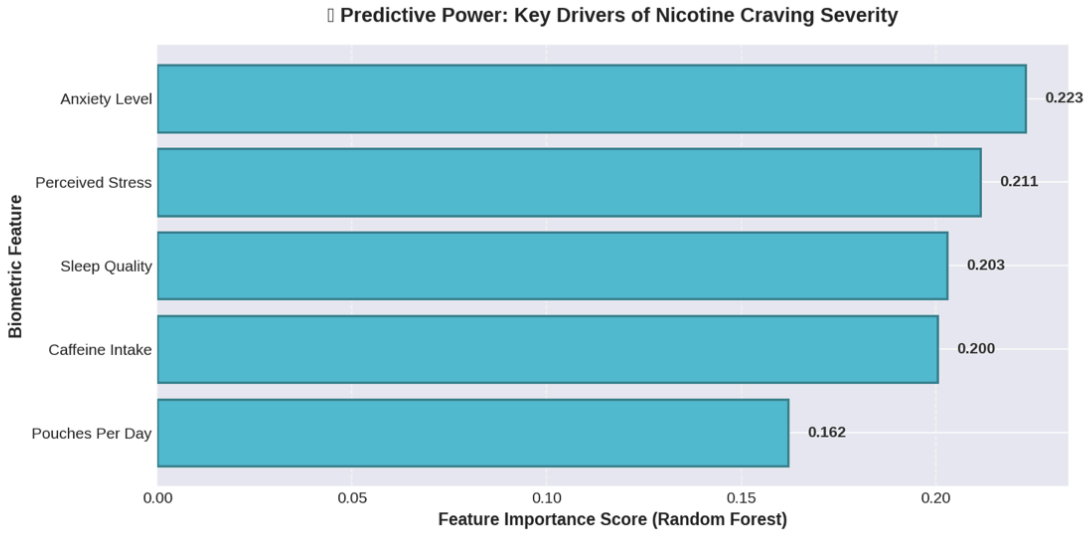
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Additionally, feature importance is evaluated using a *Random Forest Regressor*.



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Data preprocessing here involves identifying missing values, excluding records with missing values in selected features, and scaling numerical features for clustering. Unlike the first project, this one doesn't create historical features from previous-day observations.

The regression and classification tasks use a random *80/20 train-test split*, without accounting for each 30 observations belonging to separate participants. This is an important distinction from the past project, and something to keep in mind when evaluating the results.

The models used in the project include *Histogram Gradient Boosting* for regression on peak craving intensity scores, *Random Forest Classification* for predicting anxiety levels and whether they're above the median, and *K-means clustering* for identifying different groups based on behavioral factors. While I am mostly interested in the supervised learning tasks, it was interesting to see discovered patterns from K-means clustering.

The regression model is evaluated using *Mean Squared Error (MSE)*, and R^2, while the classification model is evaluated using *accuracy, precision, recall*, and *F1* scores.

The evaluation results are as follows:

```
=== Gradient Boosting Regressor Performance ===
Mean Squared Error (MSE): 2.6168
R-Squared Score (R2): 0.1301
```

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```
Classification Accuracy: 57.53%

Detailed Classification Report:
              precision    recall  f1-score   support

           0       0.57      0.62      0.59      3013
           1       0.58      0.54      0.56      2987

    accuracy                           0.58      6000
   macro avg       0.58      0.58      0.57      6000
weighted avg       0.58      0.58      0.57      6000
```

As seen above, the regression model achieves a relatively low R^2 of 0.1301, and the classification model achieved an accuracy of 57.53%, suggesting either that the selected features provide limited information for prediction, or that the selected data split is not suitable for proper evaluation of the model's performance.

I think this project is useful for its exploratory data analysis and for demonstrating how psychological and behavioral variables can be applied to different machine learning tasks. The regression experiment is especially interesting because it predicts craving scores without historical craving measurements, which makes me think about the significance of different features of this dataset for predictive performance.

Anticipated Contributions

From the information about the dataset, as well as the already existing projects, I think an interesting research question to explore is the significance of different feature groups (psychological states, lifestyle information, historical information about earlier nicotine use) for predictive performance on new participant observations. In particular, I want to implement a ***regression model*** to predict nicotine craving scores, represented in the *Overall_Craving_Score* feature. I want to compare results across multiple models that use different feature groups individually, as well as a model that combines them. My goal is to see how much the prediction accuracy changes from using different features, and whether combining features produces a meaningful change in performance, or if the prediction is accurate enough even with a limited features scope.

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Potential models that I would like to implement across the selected feature combinations would be ***Ridge Regression***, ***Random Forest***, and ***XGBoost***, judging from their performance in the existing projects I studied. I will repeat the same chronological data split as in the first project, with the training set on days 1-20, validation on days 21-25, and test set on days 26-30. For EDA, I'm thinking of including feature distributions, correlations and trends, as I found those to be the most informative. Data preparation should handle missing values and scaling, and I think it would be good to include feature engineering as well, similar to what the first project implemented. I will perform evaluation with ***MAE***, ***RMSE*** and R^2.

The main contribution is to identify whether different feature configurations (increasing from individual feature groups to different combinations of groups, and then to all features together) lead to improved performance for prediction of overall nicotine craving scores on future observations, and how significant is the difference in prediction accuracy, compared across up to three different models.